

Sachin Pemmadi

Wichita, KS | 316-372-6926 | sachinpemmadi05@gmail.com | LinkedIn: linkedin.com/in/pemmadi-sachin

Key Wins: Built RAG pipelines & AI agents with LangChain · 15% model accuracy improvement · Deployed LLM-powered features via REST APIs

PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building and deploying Generative AI applications using Python, LangChain, and Retrieval-Augmented Generation (RAG), alongside predictive ML models in TensorFlow and PyTorch that improved accuracy by 15% and targeting precision by 20%. Experienced integrating LLMs and AI agents into production systems via Django/Flask REST APIs, with strong grounding in prompt engineering, vector-based retrieval, and end-to-end ML pipelines. Seeking to apply GenAI and software engineering expertise to build production-grade AI-powered applications.

EDUCATION

Wichita State University — M.S., Data Science Engineering

Wichita, KS | May 2026

Amrita Sai Institute of Science and Technology — B.Tech., Computer Science Engineering

Paritala, India | Jun 2024

TECHNICAL SKILLS

AI / GenAI: TensorFlow, PyTorch, LangChain, Retrieval-Augmented Generation (RAG), AI Agents, Prompt Engineering, NLP, Large Language Models

Programming: Python, Java, JavaScript, SQL

ML & Data Science: Scikit-learn, Pandas, NumPy, Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, SMOTE, Time-Series Forecasting, Feature Engineering, Model Evaluation (Precision, Recall, F1, ROC-AUC, RMSE), Model Monitoring & Drift Detection

Data Engineering & Cloud: ELT, Batch Processing, Microsoft Azure AI, Databricks, Snowflake, Google BigQuery, Distributed Systems, Cloud-native Architectures

Web & Databases: Django, Flask, RESTful APIs, MySQL

Visualization & BI: Power BI, Tableau, Matplotlib, Interactive Dashboards

Tools: Docker, Git, Jupyter, PyCharm, Agile/SDLC

PROFESSIONAL EXPERIENCE

Wichita State University — Graduate Assistant, Data Science & Digital Twin Analytics Wichita, KS | Apr 2025 - Jul 2025

- Improved simulation-based model training throughput by building Python/pandas data pipelines that scaled data extraction and transformation.
- Increased predictive model accuracy by 15% by building and validating models linking CAD-backed design parameters to performance outcomes.
- Ensured reliable inputs for downstream modeling by cleaning and validating CAD-backed simulation datasets.
- Improved reproducibility and team collaboration by documenting model development and monitoring workflows in Git and Jupyter Notebook.

Brain O Vision — AI & Full-Stack Developer Intern Vijayawada, India | Jan 2024 - Apr 2024

- Enabled real-time inference for analytics pipelines by deploying ML models into production via RESTful APIs built in Django and Flask.
- Streamlined model training and inference workflows by optimizing MySQL-backed datasets and queries.
- Improved stakeholder visibility into AI-driven insights by building interactive dashboards in Tableau and JavaScript.
- Delivered data-driven features by integrating ML models into user-facing web applications with cross-functional teams.

Spypro Security Solutions — Machine Learning & Data Science Intern Vijayawada, India | Jan 2023 - Dec 2023

- Improved targeting accuracy by 20% by engineering and evaluating logistic regression, decision tree, and random forest models in scikit-learn.
- Guided business decisions by monitoring model performance (precision, recall, ROC-AUC) and presenting results to stakeholders.
- Informed marketing strategy and customer segmentation by translating business questions into predictive analytics solutions.

- Ensured data integrity for model development by extracting and preparing large-scale training datasets with Python and pandas.

Spypro Security Solutions — Cybersecurity Data Analytics Intern *Vijayawada, India | Jul 2022 - Dec 2022*

- Improved model-training accuracy by collecting and transforming large-scale network event data with Python and SQL to uncover key traffic patterns.
- Supported automated anomaly-detection evaluation by building Python monitoring scripts to track authentication behavior and flag anomalies.
- Reduced potential security incidents by analyzing large-scale network/security datasets and delivering insights that guided risk-management decisions.
- Validated detection models using statistical techniques and delivered interactive dashboards summarizing performance to stakeholders.

ACADEMIC & PERSONAL PROJECTS

Forest Fires Prediction and Analysis

- Built scikit-learn models to predict wildfire risk from environmental/meteorological data, supporting resource allocation decisions.
- Improved model performance over baseline via feature engineering, hyperparameter tuning, and cross-validation pipelines.
- Enabled real-time risk monitoring by deploying model outputs to a Power BI dashboard, with Python-based drift monitoring in production.

Stock Price Prediction with Supervised ML

- Built batch/real-time ETL pipelines in Python to engineer features from historical OHLCV data into model-ready datasets.
- Selected top-performing model by comparing regression and tree-based approaches on RMSE, R-squared, and MAE.
- Maintained live prediction accuracy by deploying the model on Azure Databricks with automated drift detection to trigger retraining.

Telecom Customer Churn Prediction (Ongoing)

- Building classification models to identify at-risk telecom customers from contract, usage, and billing data.
- Addressing class imbalance with SMOTE, optimizing for recall on the minority churn class.
- Evaluating models on precision, recall, F1, and ROC-AUC to derive retention strategies from feature importance.

Electric Vehicle Adoption Analytics

- Surfaced regional EV adoption trends by analyzing 216,000+ EV registration records across vehicle types.
- Built geographic heatmaps and time-series comparisons of BEV vs. PHEV adoption patterns.
- Highlighted affordability gaps informing EV ecosystem planning by correlating price against electric range.

CERTIFICATIONS

- Python for Data Science and Machine Learning Essential Training, Part 1 — LinkedIn Learning
- Data Science Foundations: Fundamentals — LinkedIn Learning
- Learning SQL Programming — LinkedIn Learning